

BillieBot Hardware Design and Configuration Report

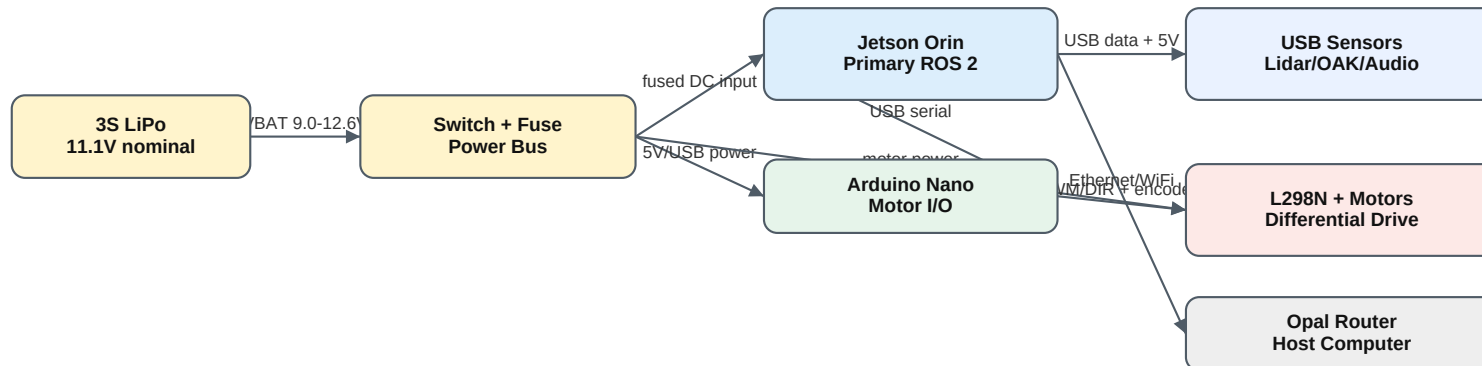
MVP build wiring, subsystem interfaces, signal map, and assembly instructions

Prepared for the BillieBot MVP: a ROS 2 differential-drive robot for autonomous apartment navigation, dog detection, bark/loud-noise detection, state logging, and daily activity summarization.

Scope note: this report intentionally does not define the Arduino Nano pin-level map. The Arduino-facing signal categories are included, but exact Nano pins for motor PWM, motor direction, encoder channels, and any interrupt assignments are left to your existing motor-controller design.

Revision: v0.1 build report | Date: July 1, 2026

Top-Level MVP Electrical/Data Architecture



Exact Arduino Nano pins are omitted by design; only signal classes are shown.

1. Executive Summary

The recommended MVP wiring architecture is a single-primary-computer robot: the Jetson Orin Nano runs the ROS 2 graph, navigation, perception, logging, and summary generation. The Arduino Nano is a dedicated low-level motor/encoder controller. The Raspberry Pi 5 is optional and should be treated as an auxiliary sensor computer unless you specifically need Raspberry Pi CSI cameras, I2C thermal sensing, or I2S audio experiments.

The practical build splits BillieBot into six electrical/data subsystems: power distribution, primary compute/networking, low-level drive control, navigation sensors, visual perception sensors, and audio I/O. Every subsystem must share a common ground reference where electrical signals cross between boards, especially across the Arduino, L298N, encoders, and battery-powered motor branch.

Critical build decisions

- Use the Jetson barrel/DC input for robot power; do not power the Jetson from its USB-C connector. The Jetson Orin Nano developer-kit USB-C port is for data/device mode, not main power.
- Use the OAK-D Lite directly on a Jetson USB 3 Type-A port using a USB-A-to-USB-C cable. It carries both 5 V power and RGB/depth/AI data.
- Use the RPLidar A1 through its USB adapter. Internally the lidar is UART, but the development adapter presents a USB serial device to the Jetson.
- Use the ReSpeaker XVF3800 through USB for MVP bark/loud-noise sensing. Keep the SPH0645 I2S mic and MAX98357A speaker amplifier as optional experiments.
- Use the L298N only as a bring-up motor driver. It is electrically simple, but inefficient and heat-prone. Upgrade to a modern MOSFET driver with current limiting before extended autonomous testing around Billie.
- Add a LiPo low-voltage cutoff/alarm. A 3S LiPo can physically run the robot, but the system should stop before the pack approaches damaging cell voltage.

Recommended MVP configuration

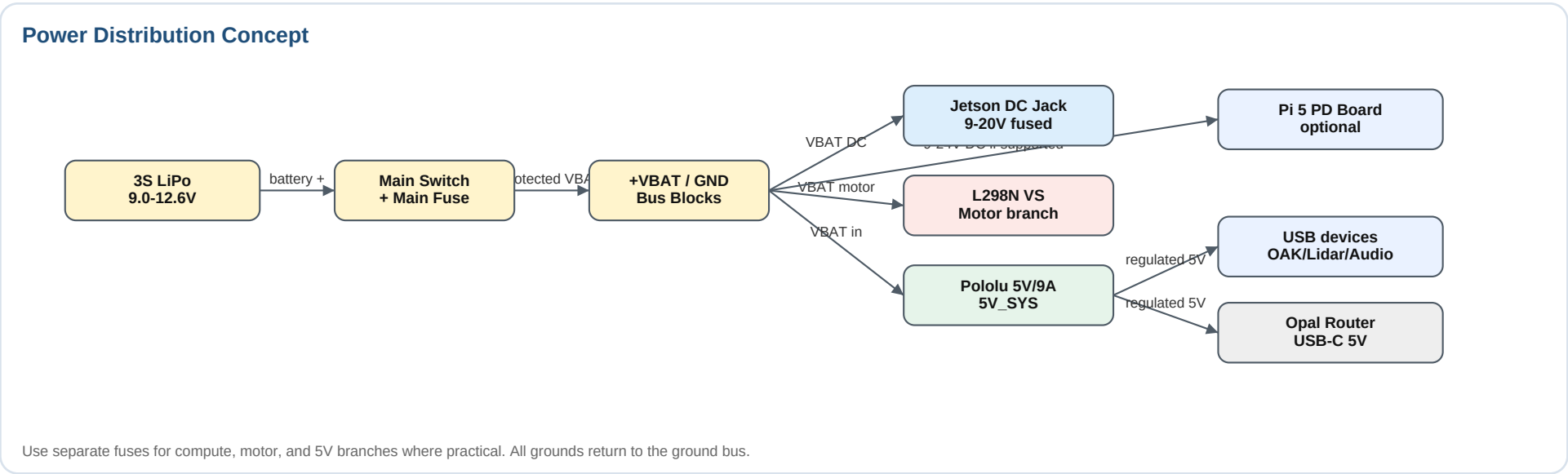
Decision	Recommended MVP choice	Reason
Primary computer	Jetson Orin Nano Super	Best match for ROS 2, Nav2, OAK-D perception, YOLO-class detection, logging, and summary generation.
Auxiliary computer	Raspberry Pi 5 optional	Use only for Pi CSI cameras, MLX90640, or I2S audio experiments after Jetson-only MVP is stable.
Main vision sensor	OAK-D Lite on Jetson USB 3	Single cable provides power and data; supports RGB/depth perception.
Main audio sensor	ReSpeaker XVF3800 on Jetson USB	USB audio minimizes I2S/Linux driver bring-up risk.
Navigation sensor	RPLidar A1 on Jetson USB serial	Simple ROS integration and 2D planar map/navigation support.
Motor controller	Arduino Nano serial bridge	Separates hard real-time-ish encoder/motor I/O from ROS 2 compute.

Source basis

The interface and power details in this report are based on the component list you provided plus vendor documentation for the Jetson Orin Nano, Raspberry Pi 5, RPLidar A1, DFRobot IMU, OAK-D Lite, Adafruit sensors/audio boards, ReSpeaker XVF3800, Pololu regulator, DFRobot GS2678 regulator, L298N, and GL.iNet Opal. A reference list appears at the end of the report.

2. Power Subsystem Design

BillieBot should use a star-like distribution from the battery through a main switch and main fuse into a positive bus and ground bus. Keep high-current motor wiring physically separate from low-level sensor and audio wiring where possible. Tie grounds at the power distribution ground bus so signal references are common but motor current does not flow through signal harnesses.



2.1 Power rails

Rail	Nominal voltage	Source	Loads	Signal carried	Build notes
VBAT	3S LiPo: 11.1 V nominal, about 12.6 V full	Venom 3S LiPo through main switch and fuse	Motor driver VS, Jetson DC input if configured, buck-regulator inputs	Unregulated DC power	Use short/high-current wiring. Add low-voltage alarm/cutoff. Do not connect directly to 5 V devices.
MOTOR_PWR	Same as VBAT	Motor branch fuse from VBAT bus	L298N motor supply input	High-current motor DC power	Keep motor power wiring twisted/short. Expect noise. Keep away from IMU/audio.
5V_SYS	5.0 V regulated	Pololu D24V90F5 from VBAT	Router, powered USB hub if used, optional 5 V sensor branch, possibly Arduino if not USB powered	Regulated DC power	Do not back-feed host USB ports. Use only if wiring path is clear. Pololu regulator supports high-current 5 V loads.
5V_PI	5.1 V typical	GeeekPi/52Pi-style Pi 5 PD board or official Pi supply	Raspberry Pi 5	Regulated Pi power	Use the PD board only per its exact input method. Some boards accept 9-24 V DC; verify your board revision.
JETSON_IN	9-20 V accepted at barrel jack per carrier-board spec	Fused VBAT or dedicated DC regulator	Jetson Orin Nano carrier board	Main Jetson power	Use center-positive 5.5 x 2.5 mm barrel plug. Do not use Jetson USB-C for power.
3V3_LOGIC	3.3 V	Jetson/Pi header regulator	I2C/I2S logic reference only	Low-current logic power	Do not use 3.3 V header as a general sensor power rail unless current is known to be small.

2.2 Fuse guidance

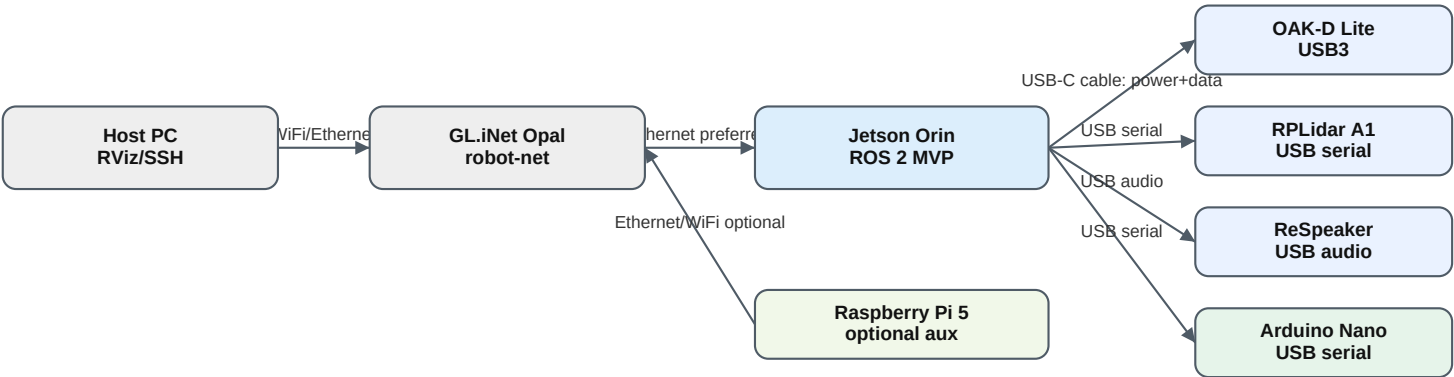
Branch	Suggested starting fuse	Reason / adjustment
Main battery	10 A automotive blade fuse	Reasonable initial protection for low-speed bring-up. Increase only after measured current requires it.
Motor branch	5 A	L298N and small gearmotors should not be allowed to draw high sustained current. If nuisance trips occur, measure stall/running current before increasing.
Jetson branch	3 A to 5 A	At 9-12 V input, a 25 W Jetson load can draw about 2-3 A before conversion losses. Use wiring and connector rated accordingly.
5V_SYS branch	5 A to 10 A depending on loads	Protect the 5 V regulator output and downstream wiring. Size to wire gauge and actual loads, not regulator maximum.
Router/Pi auxiliary branch	2 A to 5 A	Use separate branch protection if the auxiliary computer/router is powered from robot battery.

Warning: LiPo batteries can source very high fault current. Install the fuse as close to the battery positive terminal as physically practical. Never work on downstream wiring with the battery connected. Use a real LiPo charger and storage practices.

3. Compute, Networking, and USB Topology

The Jetson should be the ROS 2 master computer for the MVP. Connect high-bandwidth sensors directly to Jetson USB ports first. Add the Adafruit USB Mini Hub only for low-bandwidth devices or convenience after each device has been verified directly.

Recommended MVP Data Topology



Prefer Jetson Ethernet to the Opal LAN port for stable remote development and ROS network behavior.

3.1 Computer and network connections

Component	Connects to	Physical interface	Signal carried	Power carried	Notes
Jetson Orin Nano Super	Battery/compute power branch	5.5 x 2.5 mm center-positive barrel jack	None; power only	9-20 V DC input	Use fused DC branch. Jetson USB-C is for data/device mode, not primary power.
Jetson Orin Nano Super	Host network via Opal	RJ45 Ethernet from Jetson GbE to Opal LAN	IP network: SSH, ROS 2 DDS, file transfer, web dashboard	No PoE unless separately implemented	Ethernet is preferred over Jetson WiFi for repeatability. Configure static DHCP reservation if possible.
Jetson NVMe SSD	Jetson M.2 Key-M 2280 slot	M.2 internal connector	PCIe storage data	M.2 slot power	Already internal to Jetson. Install with thermal pad; verify it appears as NVMe storage.
Raspberry Pi 5	Pi 5 PD power expansion board	USB-C/PD or expansion-board power path	Power only	5 V / high-current Pi power	Optional. Use as auxiliary only after Jetson-only MVP works.
Raspberry Pi 5	Opal router or Jetson network	Ethernet or WiFi	ROS 2 network, sensor bridge data	No power	Use only if Pi hosts cameras/I2C/I2S devices.
GL.iNet Opal	Battery 5 V branch	USB-C power input	Power only	5 V/3 A recommended	Power from regulated 5V_SYS with a USB-C power cable, or use its wall supply during bench bring-up.
Host computer	Opal router	WiFi to robot-net or Ethernet to LAN port	SSH, ROS visualization, RViz, rosbag transfer	No robot power	Keep host off motor-power branch.

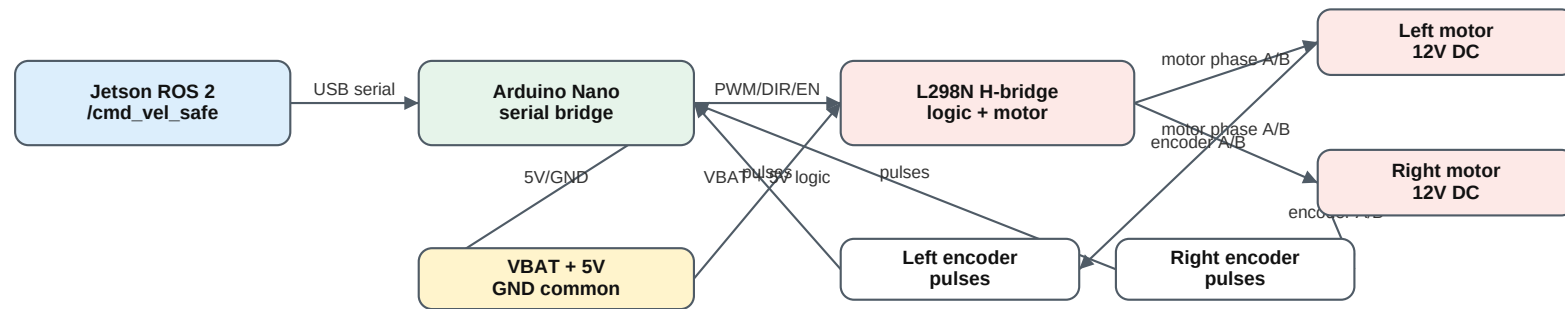
3.2 USB port assignment

Jetson port	Recommended device	Signal	Why
USB 3 Type-A #1	OAK-D Lite	USB 3 vision data + 5 V power	Highest bandwidth peripheral; avoid hubs initially.
USB 3 Type-A #2	RPLidar A1 USB adapter	USB serial lidar scan data + 5 V power	Navigation-critical; verify stable scan rate.
USB 3 Type-A #3	ReSpeaker XVF3800	USB audio + 5 V power	Primary audio sensor; may also expose control/status interfaces.
USB 3 Type-A #4	Arduino Nano	USB serial motor commands/encoder telemetry + 5 V	Drive interface; low bandwidth but timing-critical.
Adafruit USB Mini Hub	Optional low-bandwidth expansion	USB data + 5 V downstream	Use later for keyboard/mouse, auxiliary serial devices, or non-critical sensors. Avoid first-pass OAK-D through hub.

4. Drive Subsystem Wiring

The drive subsystem is split into high-current motor power and low-level control/telemetry. The Arduino Nano handles motor PWM/direction commands and encoder measurements. The L298N switches the motor current. The Jetson communicates with the Arduino over USB serial. Exact Arduino pins are intentionally omitted.

Drive Subsystem Signal and Power Flow



Exact Arduino pin numbers are intentionally omitted. Encoder and motor wires should be routed separately where possible.

4.1 Drive interconnect table

From	To	Physical interface	Signal carried	Power carried	Build notes
Jetson USB Type-A	Arduino Nano Mini-USB	USB A-to-Mini-B cable	USB serial: velocity commands, encoder/odom telemetry, stop/watchdog commands	5 V from Jetson USB unless externally powered	This is the ROS-to-motor-control bridge. Avoid loose USB cables; strain-relieve at both ends.
Arduino Nano	L298N logic inputs	Jumper wires/header harness	PWM, direction, enable, optional brake/standby	Logic-level signals only	Exact Arduino pins omitted. Keep wires short. Common ground required.
L298N OUT1/OUT2	Left JGA25-371 motor	Two motor wires	H-bridge switched motor voltage	High-current bidirectional motor power	Swap wires if motor direction is inverted relative to ROS convention.
L298N OUT3/OUT4	Right JGA25-371 motor	Two motor wires	H-bridge switched motor voltage	High-current bidirectional motor power	Swap wires if needed; verify left/right sign independently.
Motor encoders	Arduino Nano	Encoder harness	Quadrature encoder A/B pulses, encoder power, encoder ground	Usually 5 V or 3.3 V depending on encoder; verify motor datasheet	Exact Nano pins omitted. Prefer interrupt-capable pins if firmware expects them.
VBAT motor branch	L298N motor supply VS	Screw terminal or high-current terminal	DC motor supply	9-12.6 V unregulated battery power	Fuse motor branch. L298N voltage drop reduces motor voltage; expect heat.
5V_SYS or Arduino 5V	L298N logic supply	Header/screw terminal depending on module	Logic power	5 V regulated	Use one clean 5 V logic source. Do not rely on noisy motor regulator unless tested.

From	To	Physical interface	Signal carried	Power carried	Build notes
Ground bus	Jetson/Arduino/L298N/encoder grounds	Ground harness	Electrical reference	Return current	Mandatory common ground for signals. Keep motor return current out of sensor ground wiring if possible.

4.2 Mechanical drive configuration

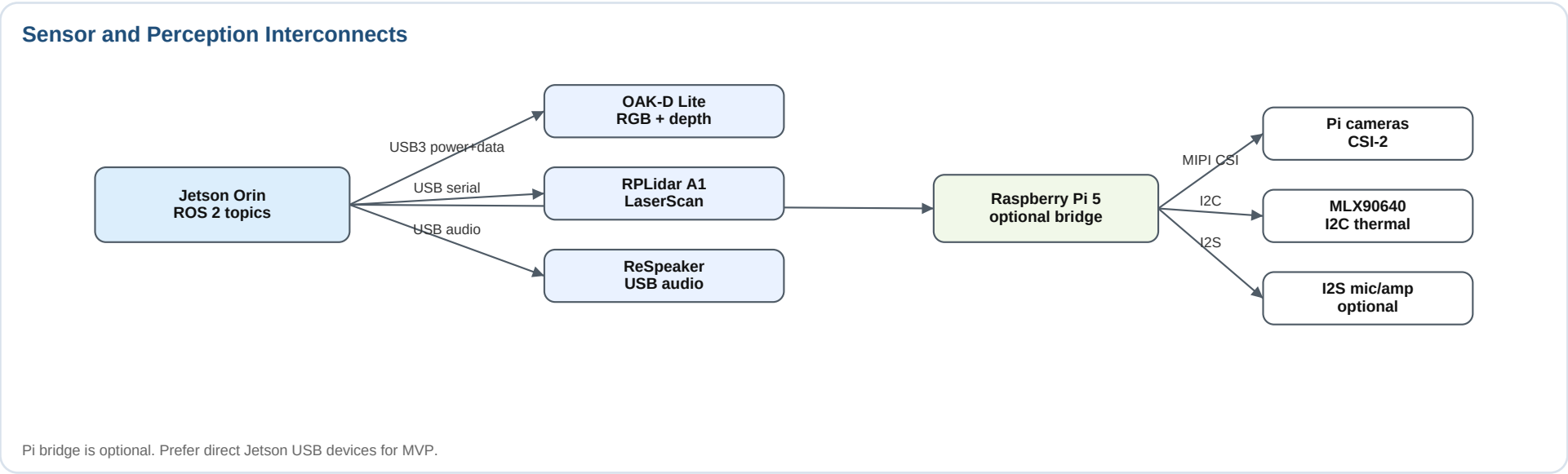
- Mount the two 68 mm wheels coaxially with the wheel axle perpendicular to the robot forward direction.
- Measure actual wheel diameter under load and actual wheel separation between contact patches; use those values in ROS parameters.
- Place the caster opposite the heavier battery/compute mass so the robot remains stable but the drive wheels retain traction.
- Add foam bumpers before testing near Billie. Keep maximum speed low until Nav2 and safety-stop behavior are verified.

4.3 Drive bring-up checks

Check	Method	Pass condition
Motor direction	Command low PWM forward with wheels lifted.	Left and right wheels rotate to drive robot forward.
Encoder sign	Rotate each wheel forward by hand or low PWM.	Encoder counts increase consistently with forward wheel motion.
1 m linear odometry	Command slow forward motion and measure floor distance.	ROS odom distance is within calibration target after adjusting wheel radius/ticks-per-rev.
360 deg rotation	Command slow in-place rotation and measure yaw.	ROS odom yaw is within calibration target after adjusting wheel separation.

5. Navigation and Perception Sensor Wiring

For MVP reliability, connect navigation and perception sensors directly to the Jetson wherever possible. The OAK-D Lite, RPLidar, and ReSpeaker are all single-cable USB peripherals from the Jetson perspective. The I2C and I2S sensors are useful but should be staged after the USB-based MVP works.



5.1 Navigation sensors

Component	Connects to	Physical interface	Signal carried	Power carried	Notes
RPLidar A1	Jetson USB Type-A	RPLidar development-board USB adapter	USB serial packets converted from internal UART; ROS publishes LaserScan	5 V USB power	Mount level, centered, and unobstructed. Use direct USB during first bring-up.
BNO055/BMP280 IMU	Jetson or Raspberry Pi I2C header	Gravity-I2C/JST-to-header harness	I2C digital sensor data: orientation, angular velocity/accel-derived data, barometer/temperature	3.3-5 V module supply; 3.3 V logic preferred	Mount rigidly to chassis. Avoid magnetic interference. Indoor magnetometer heading may be unreliable.
Wheel encoders	Arduino Nano	Motor encoder harness	Quadrature pulses	Encoder supply + ground	Pin-level details intentionally omitted. Encoder polarity/sign must be calibrated.

5.2 Visual sensors

Component	Recommended connection	Physical interface	Signal carried	Power carried	Configuration guidance
Luxonis OAK-D Lite	Jetson USB 3 Type-A	USB-A-to-USB-C cable	RGB image, stereo/depth data, device control, optional neural inference metadata	5 V from USB	Primary MVP camera. Use direct Jetson USB port; avoid low-quality hubs/cables.
Raspberry Pi Camera Module v2	Raspberry Pi 5 CSI port	FPC camera cable; Pi 5 requires 22-pin mini to standard 15-pin adapter cable	MIPI CSI-2 image data	Camera power through CSI cable	Optional. Do not use in MVP unless OAK-D is unavailable or a Pi camera experiment is needed.

Component	Recommended connection	Physical interface	Signal carried	Power carried	Configuration guidance
Raspberry Pi Camera Module 3 NoIR	Raspberry Pi 5 CSI port	FPC camera cable; connector/adapter depends on cable variant	MIPI CSI-2 image data	Camera power through CSI cable	Optional low-light/NoIR experiment. Add IR illumination only if eye/animal-safe and tested.
Adafruit MLX90640 55 deg	Raspberry Pi or Jetson I2C	STEMMA QT/Qwiic or header I2C	I2C thermal frame data	3.3-5 V supply to breakout	Auxiliary thermal sensor. Low resolution; good for warm-body/resting experiments, not primary dog detection.

5.3 Audio I/O

Component	Recommended connection	Physical interface	Signal carried	Power carried	Configuration guidance
ReSpeaker XVF3800	Jetson USB Type-A	USB cable	USB audio stream, VAD/DoA/control features depending on driver	USB 5 V	Primary bark/loud-noise sensor for MVP. Mount away from motor driver and fan airflow.
Adafruit SPH0645LM4H I2S mic	Raspberry Pi or Jetson I2S header	Header wires	I2S digital microphone data: BCLK, LRCLK/WS, data	1.6-3.3 V only	Optional. Do not connect to 5 V logic. Use only after USB ReSpeaker path works.
Adafruit MAX98357A amp	Raspberry Pi or Jetson I2S header	Header wires plus speaker terminal	I2S digital audio input; amplified speaker output	2.7-5.5 V amp power; 5 V typical	Optional future output. Not needed for MVP unless playing status tones.
Adafruit 3 W 4 ohm speaker	MAX98357A or ReSpeaker speaker output	Two speaker wires	Amplified analog/switching audio	Driven by amplifier	Use strain relief. Keep speaker wires away from microphone cabling.

6. Component-by-Component Interface Matrix

This matrix is intended as a build checklist. It describes what each component connects to, the physical interface, what signal crosses the interface, and what power is present. Arduino Nano pin numbers are not specified, but Arduino-facing signal types are included.

Component	Connects to	Interface / connector	Signal carried	Power carried	Build instruction / note
3S LiPo battery	Main switch input	Battery lead/connector to switch harness	None	VBAT: 9.0-12.6 V DC high current	Keep unplugged while wiring. Add main fuse close to battery positive.
Main on/off switch	Battery positive and main fuse	Series high-current switch	None	Battery positive switching	Switch only the positive line; ground remains common.
Main fuse	+VBAT bus	Inline blade fuse holder	None	Protected battery positive	Start with 10 A fuse for bring-up.
Copper terminal/bus blocks	All power branches	Screw terminal distribution	None	VBAT, GND, 5V_SYS depending on bus	Physically label buses. Keep VBAT and 5 V buses separate.
Pololu D24V90F5	VBAT input; 5V_SYS output	Screw/solder terminals	None	Input VBAT; output 5 V up to high-current rating	Use for regulated 5 V accessories/router/hub. Confirm polarity before connecting loads.
DFRobot GS2678 buck	VBAT input; optional auxiliary output	Screw terminals / board headers	None	Input VBAT; adjustable buck output	Optional. Because it is buck-only, it cannot hold 12 V output when 3S pack voltage falls below dropout.
Jetson Orin Nano	VBAT or dedicated regulator	5.5 x 2.5 mm barrel jack	None	9-20 V DC input; center-positive	Use fused branch. Do not power through USB-C.
Jetson Orin Nano	OAK-D Lite	USB-A to USB-C	USB3 RGB/depth/control data	5 V USB	Primary camera; connect directly first.
Jetson Orin Nano	RPLidar A1 USB adapter	USB-A cable	USB serial lidar data	5 V USB	Primary lidar. Set stable device name using udev rule later.
Jetson Orin Nano	ReSpeaker XVF3800	USB-A cable	USB audio/control	5 V USB	Primary audio input.
Jetson Orin Nano	Arduino Nano	USB-A to Mini-B	USB serial commands and telemetry	5 V USB unless externally powered	Avoid USB power backfeed if Arduino is also externally powered.
Jetson Orin Nano	GL.iNet Opal	RJ45 Ethernet	IP network	None	Use Opal LAN port. Host connects to Opal WiFi/Ethernet.
Raspberry Pi 5	PD power expansion board	Pi 5 power/PD interface	Power-control only	5 V high-current Pi power	Optional; verify exact board input options before feeding from battery.
Raspberry Pi 5	Camera Module v2	Pi 5 22-pin CSI to camera 15-pin FPC	MIPI CSI image data	Camera power	Optional; requires correct camera cable adapter.
Raspberry Pi 5	Camera Module 3 NoIR	Pi 5 CSI FPC	MIPI CSI image data	Camera power	Optional low-light camera.
Raspberry Pi 5 or Jetson	MLX90640	I2C/STEMMA QT/header	I2C thermal image frames	3.3-5 V sensor power	Optional thermal sensor. Use 3.3 V logic where possible.
Raspberry Pi 5 or Jetson	BNO055/BMP280	Gravity-I2C/header	I2C IMU/barometer data	3.3-5 V module power	Mount rigidly; calibrate orientation transform.
Raspberry Pi 5 or Jetson	SPH0645 I2S mic	I2S header wiring	I2S digital audio	1.6-3.3 V only	Optional; never apply 5 V logic.
Raspberry Pi 5 or Jetson	MAX98357A amp	I2S header wiring	I2S audio stream	2.7-5.5 V amp power	Optional; use 5 V for full speaker power.
MAX98357A	3 W 4 ohm speaker	2-wire speaker terminal	Amplified mono audio	Amplified output	Optional future speech/status output.

Component	Connects to	Interface / connector	Signal carried	Power carried	Build instruction / note
L298N motor driver	VBAT motor branch	Screw terminal	None	Motor supply voltage	Fuse branch; heatsink airflow recommended.
L298N motor driver	Arduino Nano	Header/jumper harness	PWM, direction, enable signals	Logic-level reference only	Exact Nano pins omitted.
L298N motor driver	Left/right motors	Motor output screw terminals	Switched motor voltage	Bidirectional motor current	Twist motor pair wires; verify left/right direction.
Motor encoders	Arduino Nano	Encoder harness	Quadrature pulses	Encoder supply + ground	Exact Nano pins omitted; check encoder voltage before connection.

7. Build Instructions

Use this sequence to reduce integration risk. Do not install all sensors and power everything at once. Bring up one branch at a time and verify voltage, current, and ROS visibility before adding the next branch.

7.1 Mechanical and power base

- Mount the two drive motors, wheels, caster, battery, switch, fuse holder, bus blocks, L298N, Jetson, and regulator boards on a rigid chassis plate.
- Route battery positive through the main switch, then the main fuse, then to the +VBAT bus. Route battery negative directly to the GND bus.
- Create labeled branches from the bus: Jetson branch, motor branch, 5V_SYS regulator branch, optional Pi/router branch.
- Before connecting electronics, use a multimeter to verify VBAT polarity and voltage at every branch connector.
- Install foam bumpers and ensure no sharp edges, loose wires, or exposed terminals are accessible to Billie.

7.2 Compute bring-up

- Power the Jetson from its normal supply or fused battery branch. Confirm the green power LED and successful boot.
- Connect Jetson Ethernet to an Opal LAN port. Connect host computer to Opal WiFi or Ethernet. Confirm SSH to Jetson.
- Connect the NVMe SSD internally and verify it appears in Linux before using it for logs/rosbags.
- Connect USB sensors one at a time: Arduino, lidar, OAK-D, ReSpeaker. Record each /dev device path and create udev rules later.

7.3 Drive bring-up

- Lift the robot so wheels cannot touch the ground.
- Connect the Arduino to the Jetson over USB. Verify serial communication before enabling motor power.
- Connect L298N logic power and ground. Confirm common ground between Arduino, L298N, and battery ground bus.
- Connect motor branch power to L298N VS through a branch fuse.
- Command very low PWM and verify left/right wheel direction. Correct by swapping motor output wires or firmware sign parameters.
- Verify encoder count signs before allowing ground motion.

7.4 Sensor bring-up

- Mount RPLidar level and centered above the chassis. Verify /scan before running SLAM.
- Mount OAK-D Lite facing forward at a height that sees both Billie and obstacles. Verify RGB and depth streams before running detection.
- Mount ReSpeaker away from motor driver and fan exhaust. Verify microphone input with a clap/loud-noise test.
- Add IMU only after odometry and lidar are stable. Keep it away from motor magnets and high-current wires.

7.5 Optional Raspberry Pi branch

- Install the Pi 5 PD power expansion board according to its board-specific documentation.

- Feed the PD board from its supported input only. If using DC input, verify the supported input voltage range for your exact board revision.
- Connect Pi cameras to Pi CSI connectors using the correct 22-pin/15-pin FPC adapter cable.
- Connect MLX90640 and/or BNO055 to Pi I2C if you prefer to isolate I2C bring-up from the Jetson. Publish data back to ROS 2 over the robot network.

8. Pre-Power and Smoke-Test Checklist

8.1 Pre-power checklist

Item	Pass condition
Battery disconnected while wiring	No live wiring changes occur with LiPo connected.
Main fuse installed	Fuse is installed close to battery positive; correct initial rating.
Bus polarity verified	+VBAT and GND labels match measured polarity.
5V_SYS verified unloaded	Pololu output measures about 5 V before loads are attached.
Jetson barrel polarity verified	Center pin positive; voltage within Jetson input range.
No exposed conductors	All bus screws, terminals, and battery leads are insulated or enclosed.
Common ground verified	Continuity exists between Arduino GND, L298N GND, Jetson GND reference through USB/shield or explicit ground, and battery GND where signals cross.
Motor outputs isolated	Motor terminals are not shorted to chassis or logic rails.
Wheels lifted for first drive test	Robot cannot run away during first PWM test.

8.2 Electrical smoke-test order

Step	Power connected	Data connected	Expected result
1	Battery to switch/fuse/bus only	None	VBAT appears at bus; no heat/smell/current spike.
2	5V_SYS regulator only	None	5 V output stable unloaded, then with dummy load if available.
3	Jetson branch	Ethernet to Opal/host	Jetson boots; SSH works.
4	Arduino USB	USB serial to Jetson	Arduino enumerates as serial device; motor power still disabled.
5	L298N logic only	Arduino control harness	No motor movement; logic voltages stable.
6	Motor branch enabled	Arduino to L298N	Wheels spin only under explicit low-speed command.
7	RPLidar USB	USB serial	/scan publishes.
8	OAK-D USB	USB3	RGB/depth streams publish.
9	ReSpeaker USB	USB audio	Audio device appears; loud event test publishes.
10	All MVP branches	Full ROS launch	No brownouts during low-speed motion and sensor streaming.

8.3 ROS launch verification mapping

Hardware launched	Expected ROS evidence	Likely physical interface to inspect if failed
RPLidar	/scan topic with non-empty ranges	USB cable, lidar power, USB serial permissions, spin motor obstruction.
Drive	/cmd_vel_safe accepted; /odom and /tf update	USB serial to Arduino, motor branch fuse, L298N logic/motor power, encoder harness.
SLAM	/map topic appears and updates	Lidar mounting, /tf tree, odom quality.
AMCL	/amcl_pose and map->odom TF	Map file path, initial pose, /scan, /tf tree.
Nav2	Lifecycle nodes active; robot produces /cmd_vel	Costmap params, footprint, lidar, odom, localization.
OAK-D	RGB/depth image topics	USB3 cable/port, DepthAI permissions, device current.

Hardware launched	Expected ROS evidence	Likely physical interface to inspect if failed
Vision detector	/billie/detections	Image topic names, model path, detector backend.
Audio	/billie/audio_events	USB audio device, ALSA/Pulse/PipeWire device selection, threshold.
Logger	SQLite database created and rows inserted	Filesystem permissions, database path, state topic.

9. Harnessing, Mounting, and Configuration Guidance

9.1 Wiring conventions

Signal class	Suggested wire	Routing guidance
Battery/main motor power	16-18 AWG stranded for short hobby-robot runs; size larger if measured current requires	Short, fused, mechanically secured. Keep away from audio and IMU harnesses.
5 V accessory power	18-22 AWG depending on current	Bundle separately from motor outputs. Label 5 V and GND clearly.
Motor output wires	18-22 AWG depending on motor current	Twist each motor pair. Route away from IMU and mic.
Encoder/signal wires	24-28 AWG	Use twisted pairs or grouped harness; keep away from motor outputs if possible.
I2C/I2S	Short 24-28 AWG jumpers or STEMMA/Qwiic cables	Keep short. Use 3.3 V logic. Avoid parallel routing with motor wires.
USB	Known-good shielded cables	Strain-relieve. Use short USB3 cable for OAK-D.

9.2 Mounting recommendations

- Battery: low and central to reduce tipping and improve wheel traction.
- Jetson: mechanically isolated from motor vibration where possible, with airflow around heatsink/fan.
- RPLidar: top/center, level, unobstructed 360 deg field of view. Avoid wires crossing the lidar scan plane.
- OAK-D Lite: front-facing, rigidly mounted, with known transform to base_link. Slight downward pitch may help see a small dachshund near the robot.
- ReSpeaker: top/front location, away from motor driver heat and fan exhaust. Add a soft gasket or vibration isolation if motor noise couples into the chassis.
- IMU: near the robot center of rotation, rigidly mounted, away from magnets and high-current motor wires.
- L298N: leave space for heatsink airflow. Do not enclose tightly during first drive tests.

9.3 Grounding and noise control

- Use one ground bus as the primary return/reference point.
- Do not daisy-chain motor current through the Jetson, Arduino, or sensor ground wires.
- Keep the motor branch physically separate from USB/audio/I2C harnesses.
- If the lidar or OAK-D resets during motion, suspect 5 V droop, USB cable quality, or motor-noise coupling.
- If audio detections false-trigger during driving, collect audio while motors run and raise thresholds or add vibration/noise isolation.

10. Initial Hardware Configuration Values

These are initial values for bring-up. Replace with measured values before relying on autonomous navigation.

Parameter	Initial value	How to refine
Wheel diameter	68 mm nominal	Measure loaded diameter or calibrate using 1 m forward-drive test.
Wheel radius	34 mm nominal	Adjust until odom distance matches measured floor distance.
Wheel separation	Measure actual left/right wheel contact-patch spacing	Adjust until 360 deg rotation odom matches measured yaw.
Max linear speed	0.15-0.25 m/s	Increase only after Nav2, collision monitor, and manual stop are verified.
Max angular speed	0.5-0.8 rad/s	Tune for stable lidar localization and low tip risk.
Jetson input	Fused 9-20 V DC at barrel	Confirm pack voltage does not sag below allowed range under motor load.
Main fuse	10 A initial	Measure current during all-sensors + low-speed drive; adjust only if justified.
Motor branch fuse	5 A initial	Measure motor current; L298N heat/stall tests should guide changes.
5V_SYS	5.0 V	Measure at load during OAK/ReSpeaker/lidar streaming if devices are externally powered.
LiPo cutoff	About 3.4 V/cell conservative under load	Use battery alarm/cutoff; avoid deep discharge.

10.1 Recommended MVP vs optional hardware

Hardware	MVP status	Connection choice
Jetson Orin Nano	Required	Primary computer over DC power, Ethernet, USB sensors.
Arduino Nano	Required	USB serial to Jetson; motor/encoder pins per your firmware.
RPLidar A1	Required	USB adapter to Jetson.
OAK-D Lite	Required/recommended	USB3 to Jetson.
ReSpeaker XVF3800	Required/recommended	USB to Jetson.
BNO055/BMP280	Recommended after base drive works	I2C to Jetson or Pi.
Raspberry Pi 5	Optional	Auxiliary sensor bridge only.
Pi Camera v2 / Pi Camera 3 NoIR	Optional	CSI to Raspberry Pi 5.
MLX90640	Optional	I2C to Pi or Jetson.
SPH0645 I2S mic	Optional	I2S to Pi/Jetson; 3.3 V only.
MAX98357A + speaker	Optional/future	I2S to Pi/Jetson; speaker output to 4 ohm speaker.

11. References

The following references were used to verify component interface and power assumptions. Always check the exact revision of your board/module before applying power.

ID	Reference	URL
S1	NVIDIA Jetson Orin Nano Developer Kit hardware layout: USB-C data only, USB Type-A ports, DC power jack, MIPI CSI connectors, M.2 slots.	https://docs.nvidia.com/jetson/orin-nano-devkit/user-guide/latest/hardware_layout.html
S2	NVIDIA Jetson Orin Nano Super Developer Kit specifications: compute capability and 7-25 W power range.	https://www.nvidia.com/en-us/autonomous-machines/embedded-systems/jetson-orin/nano-super-developer-kit/
S3	NVIDIA forum/DevKit carrier specification discussion: Orin Nano Dev Kit barrel jack accepts 9-20 V and USB-C is not for power.	https://forums.developer.nvidia.com/t/powering-the-nano-orin/366665
S4	Raspberry Pi 27 W USB-C power supply: Pi 5 high-power peripheral power profile context.	https://www.raspberrypi.com/products/27w-power-supply/
S5	Raspberry Pi camera documentation: Camera Module 2 and Camera Module 3 specifications.	https://www.raspberrypi.com/documentation/accessories/camera.html
S6	Raspberry Pi 5 FPC camera cable/interface: two 22-way 0.5 mm MIPI connectors and 22-to-15 pin camera cables.	https://www.adafruit.com/product/5819
S7	RPLidar A1 datasheet/dev kit manual: UART interface and USB adapter power/data behavior.	https://cdn-shop.adafruit.com/product-files/4010/4010_datasheet.pdf
S8	RPLidar A1 specification summary: 5 V power, UART 115200 bps, scan behavior.	https://www.waveshare.com/wiki/RPLIDAR_A1
S9	DFRobot SEN0253 BNO055 + BMP280 IMU: 3.3-5 V operating voltage and Gravity-I2C interface.	https://wiki.dfrobot.com/sen0253/
S10	Luxonis OAK-D Lite hardware documentation: depth range and hardware details.	https://docs.luxonis.com/hardware/products/OAK-D%20Lite
S11	Luxonis OAK USB deployment guide: OAK USB cameras use USB-C for power and data.	https://docs.luxonis.com/hardware/platform/deploy/usb-deployment-guide
S12	Adafruit MLX90640 55 deg thermal camera breakout: I2C, 3.3-5 V supply, current, FOV.	https://www.adafruit.com/product/4407
S13	Seeed ReSpeaker XVF3800 introduction: USB 4-mic array, VAD/DoA/beamforming features.	https://wiki.seeedstudio.com/respeaker_xvf3800_introduction/
S14	Seeed ReSpeaker XVF3800 product/spec page: USB 5 V, external 5 V, audio output, sample rate.	https://www.seeed.cc/product/respeaker-mic-array-v2-0
S15	Adafruit SPH0645LM4H I2S mic guide: digital I2S microphone, 1.6-3.3 V only.	https://cdn-learn.adafruit.com/downloads/pdf/adafruit-i2s-mems-microphone-breakout.pdf
S16	Adafruit MAX98357A amplifier: I2S input, 2.7-5.5 V, 3.2 W into 4 ohm speaker at 5 V.	https://learn.adafruit.com/adafruit-max98357-i2s-class-d-mono-amp/overview
S17	ST L298 datasheet: dual full-bridge motor driver, supply voltage/current and thermal limitations.	https://www.st.com/resource/en/datasheet/l298.pdf
S18	DFRobot GS2678/DFR0205 buck converter: 3.6-25 V input, adjustable output, 5 A/25 W class.	https://wiki.dfrobot.com/dfrobot0205/
S19	Pololu D24V90F5 regulator: up to 38 V input, 5 V output around 9 A, efficiency range.	https://www.pololu.com/product/2866
S20	GL.iNet Opal GL-SFT1200 product specs: Type-C 5 V/3 A input, Ethernet, WiFi, power consumption.	https://www.gl-inet.com/en-us/products/gl-sft1200